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AI Transparency and Governance Protocol

Version 2026

CAPSUS

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1 Ethical Statement and Regulatory Alignment

CAPSUS integrates Artificial Intelligence (AI) to enhance analytical speed, spatial accuracy, and institutional clarity. Our AI strategy is built on the principles of explainability, human-centric design, and environmental sustainability.

This protocol ensures that our methodologies align with the European Union AI Act (2024) and the OECD AI Principles, providing a transparent framework for clients and stakeholders.

2 EU AI Act Risk Classification

Based on the nature of our consulting services, CAPSUS classifies its AI applications as follows:

- **High-Risk Systems (Annex III - Infrastructure & Essential Services):** When AI is used to prioritize the reconstruction of damaged infrastructure (e.g., Ukraine MFB project) or manage critical urban services, CAPSUS implements stringent quality controls, as these systems may impact the safety and fundamental rights of citizens.
- **Limited/Minimal Risk:** Applications such as automated satellite image classification for land use or NLP for report analysis are treated under "Transparency Obligations," ensuring users are aware of AI interaction.

3 Technical Implementation & Methodology

Our AI ecosystem utilizes industry-standard, open-source, and cloud-based tools (Google Earth Engine, Python/Google Colab, R, QGIS) to ensure reproducibil-

ity.

Key Phases of AI Support:

- **Diagnostic Phase:** Deep Learning and Machine Learning (Random Forest, XGBoost) for satellite imagery classification and land-use change modeling.
- **Institutional Analysis:** Natural Language Processing (NLP) to parse government reports, identify investment dependencies, and extract policy insights.
- **Predictive Modeling:** Advanced statistical models in Python/R to simulate climate adaptation scenarios and urban growth paths.
- **Dissemination:** AI-powered automated dashboards that translate complex data into actionable evidence for decision-makers.

4 Data Governance and Integrity

To prevent "garbage in, garbage out" and ensure non-discriminatory outcomes, CAPSUS follows a strict data lifecycle:

1. **Sourcing:** We prioritize verified public datasets and high-resolution satellite data.
2. **Bias Mitigation:** Models are tested against diverse demographic and geographic variables to ensure reconstruction or investment priorities do not marginalize vulnerable populations.
3. **Data Sovereignty:** Our tools do not feed commercial or third-party internal models. All client data remains within the project's secure environment.

5 "Human-in-the-loop" & Anti-Black Box Policy

CAPSUS rejects "black box" AI. Our Responsible AI Use framework includes:

- **Human Validation:** Every AI-generated output is reviewed by our lead consultants and geospatial analysts (e.g., specialists in urban planning and environmental science) before delivery.
- **Process Splitting:** We decompose complex AI tasks into traceable steps. This modular approach allows us to audit the logic at each stage and explain why a specific recommendation was made.
- **Traceability:** All code and modeling parameters are documented, allowing clients to replicate results using open-source tools.

6 Environmental Sustainability (Green AI)

In line with our core mission as a sustainability consultancy, we optimize our AI processes to minimize energy consumption:

- **Computational Efficiency:** We utilize cloud-based infrastructure (Google Cloud) which operates with high energy efficiency and carbon-neutral commitments.
- **Algorithm Optimization:** We prioritize "lean" models that provide high accuracy without the excessive computational cost of over-parameterized systems.

7 Accountability

This strategy aligns with the goals of improving data quality, supporting climate adaptation, and enabling evidence-based decision-making. By maintaining full transparency, CAPSUS ensures that AI serves as a tool for public good, fostering trust between governments, contractors, and the citizens they serve.